

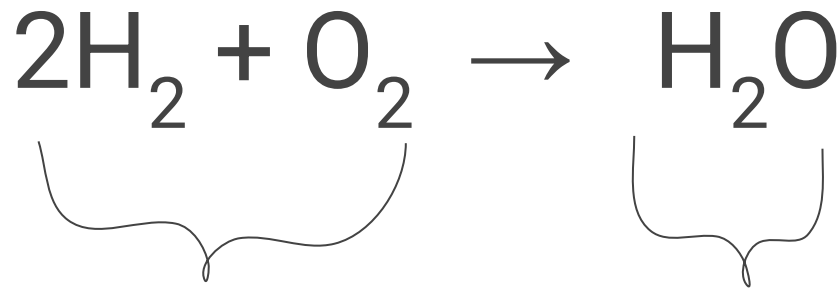
CRAM Session - Chemical Reactions

Alexander Lau, 1/25/24

MSJ STEM Success

Structure of a Chemical Equation

- Represents a chemical reaction through chemical formulas
- **Ex:** Hydrogen (H_2) + Oxygen (O_2) $\rightarrow$ Water (H_2O)
- Arrow = “Produces or forms”

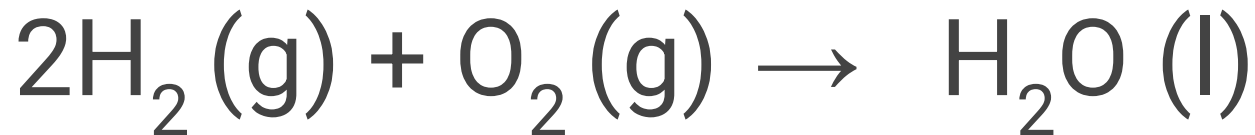


Reactants: “Before”
the arrow

Products: “After”
the arrow



Structure of a Chemical Equation



↑
—— (Gas) ——
↑

↑
Represents liquid
water

- Sometimes, chemical equations include **state of matter**
 - Solid (s)
 - Liquid (l)
 - Gas (g)
 - Aqueous (aq)

Another Example



Sodium cation



Chlorine anion

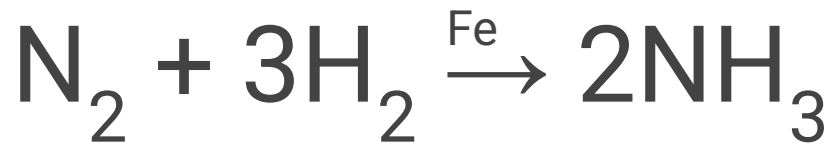


Sodium chloride

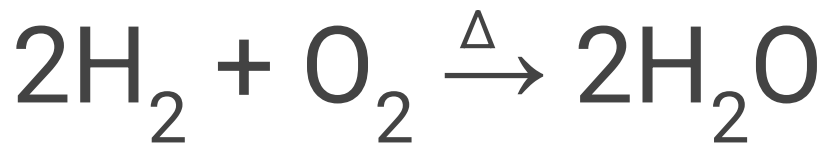
- (aq) → Aqueous (Dissolved in water)
- Cations have positive charges, anions have negative charges

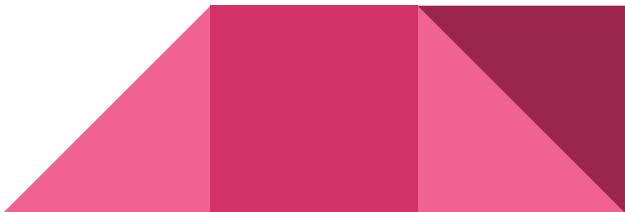


Other Symbols



- Fe (Iron) is the reaction **catalyst** (makes it faster)



- Delta (Δ) represents an input of heat
- 

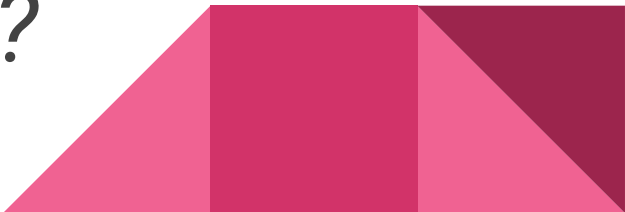
Diatomic Elements

Sodium $\rightarrow$ Na

*Calcium $\rightarrow$ Ca

But why is Hydrogen = H_2 ,

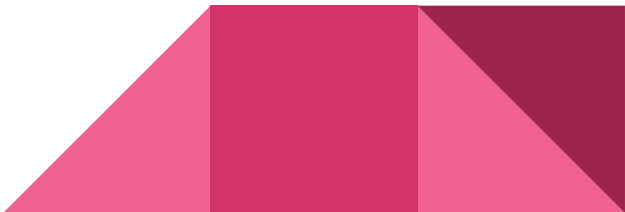
and Oxygen = O_2 ?



Diatomic Elements

- Most elements, like Na & Ca, are stable individually (can exist as individual atoms)
- But **seven different** elements prefer to exist in pairs

Hydrogen	Nitrogen	Oxygen	Fluorine	Chlorine	Bromine	Iodine
H ₂	N ₂	O ₂	F ₂	Cl ₂	Br ₂	I ₂

- Pairs make these elements more stable
- 

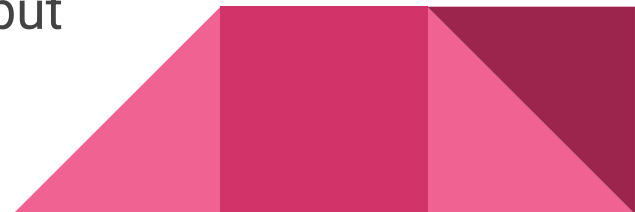
Diatomic Elements

How to Remember Diatomic Elements:

Hydrogen	Nitrogen	Fluorine	Oxygen	Iodine	Chlorine	Bromine
H_2	N_2	O_2	F_2	I_2	Cl_2	Br_2

- **Have No Fear Of Ice Cold Beer**

- I've personally never used this mnemonic, but whatever helps you memorize!



Diatomic Elements

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Period ↓																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	* 71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	* 103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
			* 57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb		
			* 89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No		

- Another Method: Remember Hydrogen (H), then trace a “7” with your finger.

Diatomic Elements

Example: Are the following elements diatomic?

13	14	15	16	17	18
					2 He
5 B	6 C	7 N	8 O	9 F	10 Ne
13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og

- 1) Bromine / Br
- 2) Phosphorus / P
- 3) Argon / Ar
- 4) Hydrogen / H
- 5) Helium / He
- 6) Astatine / At

Diatomic Elements

Example: Are the following elements diatomic?

Group → 1	15	16	17	18		
Period ↓ 1	1 H			2 He		
	5 B	6 C	7 N	8 O	9 F	10 Ne
	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og

- 1) Bromine / Br (**yes** / Br_2)
- 2) Phosphorus / P (**no**)
- 3) Argon / Ar (**no**)
- 4) Hydrogen / H (**yes** / H_2)
- 5) Helium / He (**no**)
- 6) Astatine / At (**no**)

Skeleton & Word Equations

Skeleton & Word Equations

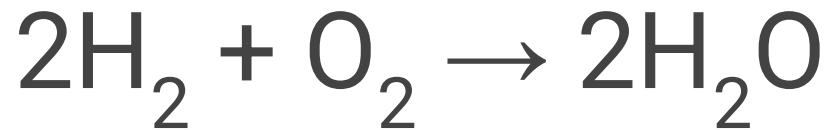
- Word Equations: Uses the names of compounds instead of formulas.
 - Remember chemical nomenclature (naming conventions)!
 - You'll need to convert the words into formulas

Hydrogen + Oxygen $\rightarrow$ Water

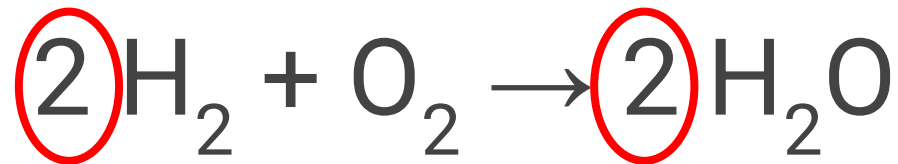


Skeleton & Word Equations

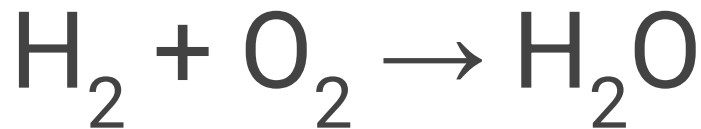
- Skeleton Equations: Show the **formulas involved**; the coefficients of the equation aren't balanced.
- **Example**: Is this a skeleton equation? Why or why not?



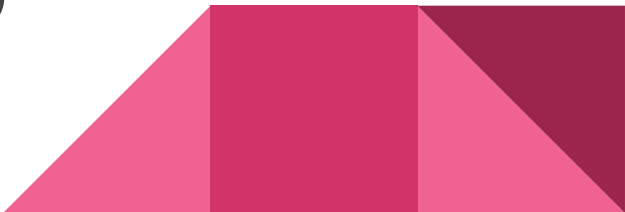
Skeleton & Word Equations



This equation has coefficients already! Therefore, it is not a skeleton equation.

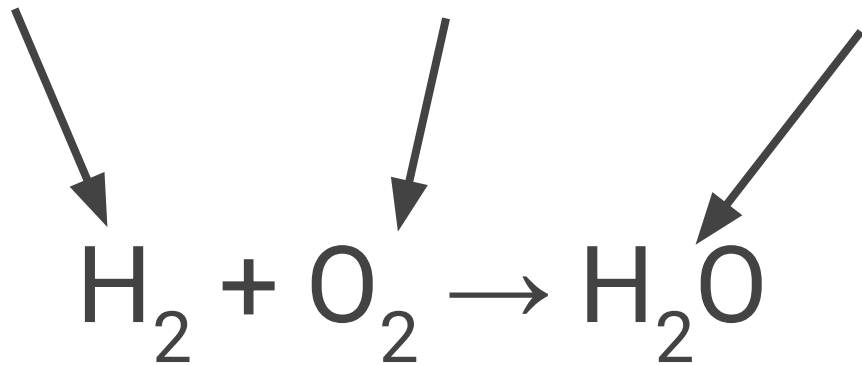


(real skeleton equation)



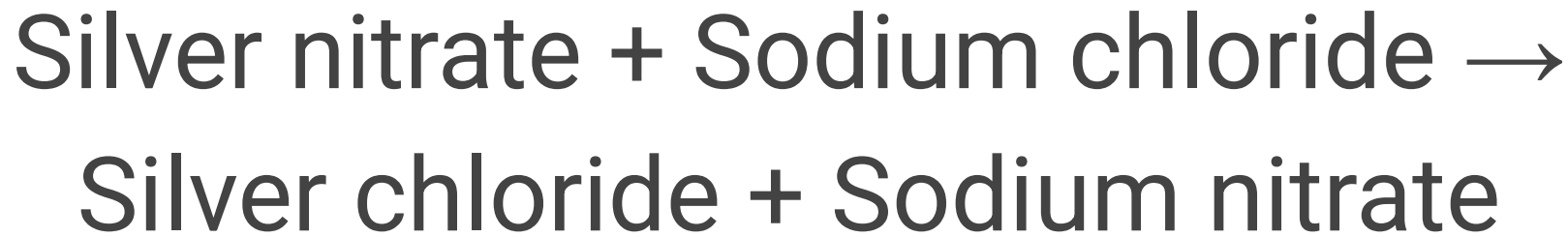
Skeleton & Word Equations

Hydrogen + Oxygen $\rightarrow$ Water



Skeleton & Word Equations

Example: Convert the following word equation into a skeleton equation.



Skeleton & Word Equations

Example: Convert the following word equation into a skeleton equation.

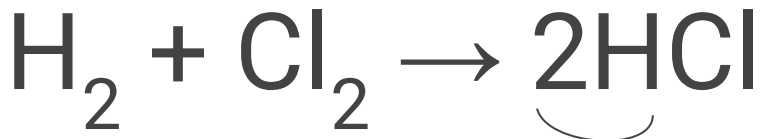
Silver nitrate + Sodium chloride →
Silver chloride + Sodium nitrate



Balancing Chemical Equations

Balancing Chemical Equations

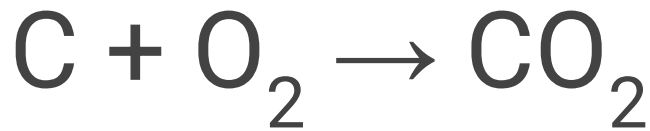
- Law of Conservation of Mass: No mass is “created” or “destroyed” during a chemical reaction.
- *For chemical equations*, count the number of atoms in the reactants and products. The equation is **unbalanced** if they’re not equal.



2 Hydrogen = 2 Hydrogen

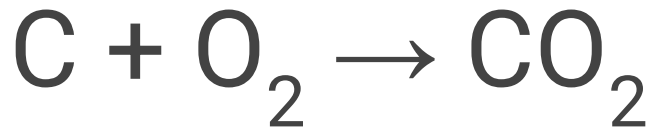
Balancing Chemical Equations

- **Example:** Look at the following equation.
 - Is this equation balanced?
 - Could this equation be considered a skeleton equation?



Balancing Chemical Equations

- **Example:** Yes, the equation is balanced. There is one Carbon atom and two Oxygen atoms in both the reactant and product sides.
- The equation can also be considered a skeleton equation! Sometimes, the equation already is balanced.



- In case you're curious: The states of matter are
 $\text{C (s)} + \text{O}_2 \text{ (g)} \rightarrow \text{CO}_2 \text{ (g)}$



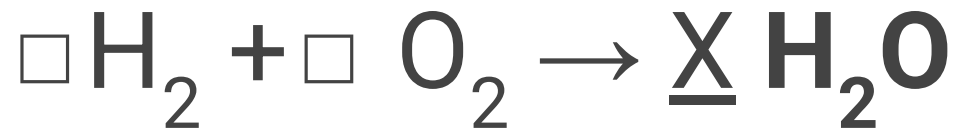
Balancing Coefficients (Simple Example)

- This *skeleton equation* isn't balanced yet.
- There are 2 Oxygen atoms in the reactants, but only one in the products!
- Clearly we need to tweak the coefficients.
 - By filling in the missing boxes



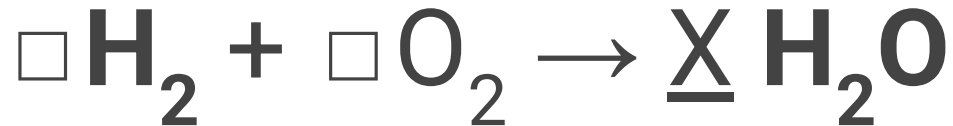
Balancing Coefficients (Simple Example)

- Kind of excessive for this problem, but first identify the most complicated molecule
 - Molecule with the most components
- Label that coefficient as x



Balancing Coefficients (Simple Example)

- Look for atoms that only appear in 1 reactant and 1 product
- Both hydrogen and oxygen meet this criteria



- H_2 and H_2O both have 2 moles of hydrogen each
- Thus, there should be an *equal coefficient* on both equations



Balancing Coefficients (Simple Example)

- For oxygen, there are 2 moles of oxygen for 1 mole of O_2
 - There are only X moles of oxygen
- Therefore, the coefficient of O_2 is $X/2$

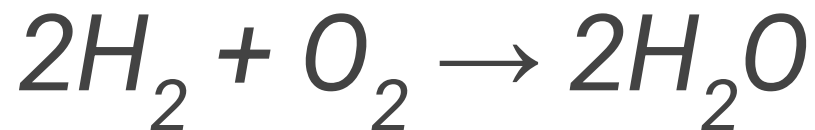


- Simplify by cancelling out X and eliminating fractions
 - Multiply all coefficients by 2



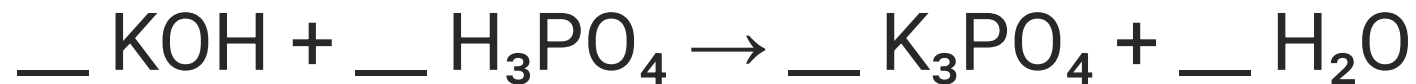
Balancing Coefficients (Simple Example)

- That gives us our answer:



Balancing Coefficients

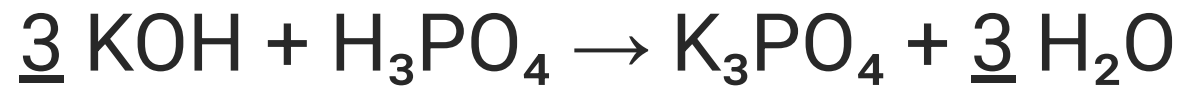
- Example: Balance this equation.



(Source: Thoughtco)



Balancing Coefficients

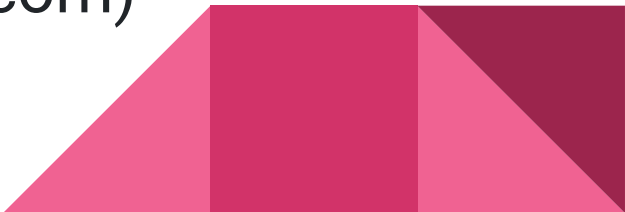


Balancing Coefficients (Hard Example)

- This *skeleton equation* isn't balanced yet.
- For example, there is 1 Chlorine atom in the reactants, but four in the products.
- Clearly we need to tweak the coefficients.
 - Fill in the missing boxes



(Source: equationbalancer.com)



Balancing Coefficients

- Pick the most “difficult” molecule to begin with.
 - Preferably one with multiple components
- Label the coefficient “X.”



Balancing Coefficients

- Look for atoms that only appear **once** in both the reactant and product sides.
 - Here it's Potassium (K).



- Both KMnO_4 and KCl have **one** potassium ion naturally.
 - To maintain this ratio, KCl 's coefficient is also **X**.
- 

Balancing Coefficients

- Fill in “X” for KCl’s coefficient.



- Next, Manganese (Mn) only appears once in KMnO_4 and MnCl_2 .
- Therefore, MnCl_2 also has a coefficient of X.



Balancing Coefficients

- Oxygen only appears in KMnO_4 and H_2O .



- However, KMnO_4 has **four** oxygen atoms;
 H_2O only has **one**.
- This means that **four times** the oxygen molecules are required to balance H_2O with KMnO_4 .



Balancing Coefficients

- Therefore, H₂O's coefficient is 4X.



Balancing Coefficients

- HCl only has one Hydrogen atom; H₂O has two.
- Therefore, there are **double** the moles of HCl as H₂O.
- That makes HCl's coefficient (4X) * 2 = 8X.



Balancing Coefficients

- Finally, for Cl, we know that there will be $8X$ moles of Cl in the reactant side (based on HCl).
- However, the product side already has $2X$ moles of Cl from MnCl_2 and X moles of Cl from KCl.
 - 1 mole of MnCl_2 has 2 moles of Cl.



- That leaves $5X$ moles of Cl.
- 

Balancing Coefficients

- There is 1 mole of Cl_2 for every 2 moles of Cl.
- Therefore, the coefficient for Cl_2 is $5X / 2$.



Balancing Coefficients

- Divide every coefficient by X; X is just a placeholder variable.



- Multiply out fractions.



- And we have our answer!



Types of Chemical Reactions

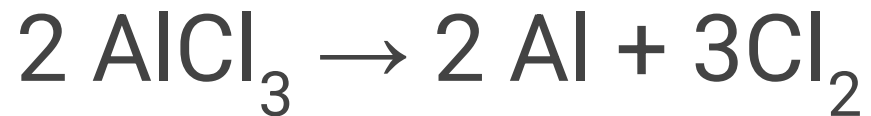
Synthesis Reactions

- 2 reactants “combine” to form a single product
- General pattern: $A + B \rightarrow AB$



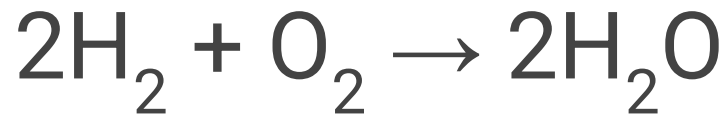
Decomposition Reactions

- One compound breaks down into 2+ different products
 - Typically requires heat to break down
- $AB \rightarrow A + B$



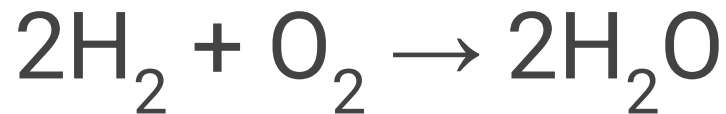
Example

Is this a *synthesis* reaction or a *decomposition* reaction?



Example

Is this a *synthesis* reaction or a *decomposition* reaction?

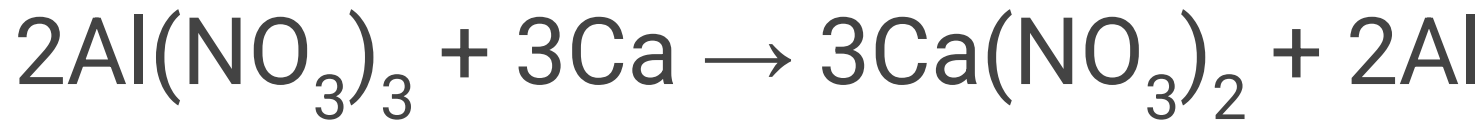


Multiple reactants form one product, so this is a synthesis reaction.



Single Replacement Reactions

- AKA 'single displacement'
- Involves 1 compound and 1 element
- The element "displaces" another element already in the compound and takes its spot
- $AB + X \rightarrow XB + A$



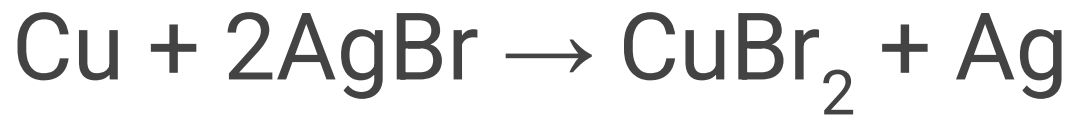
Double Replacement Reactions

- AKA 'double displacement'
- Involves two ionic compounds that swap cations (positive ions, e.g. sodium and silver) to form two new ionic compounds
- $AB + XY \rightarrow XB + AY$



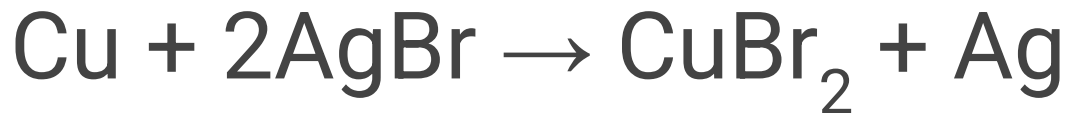
Example

Is this a *single displacement* reaction or a *double displacement* reaction?



Example

Is this a *single displacement* reaction or a *double displacement* reaction?



There is one compound and one individual element in both the reactants and products; therefore, this is a *single displacement* reaction.



Combustion

- Special type of reaction
- Includes a source of heat & oxygen (O_2) as a reactant
- Substance being combusted $\rightarrow$ Typically organic
 - Only made of C, H, and O
- CO_2 and H_2O are generally the only products



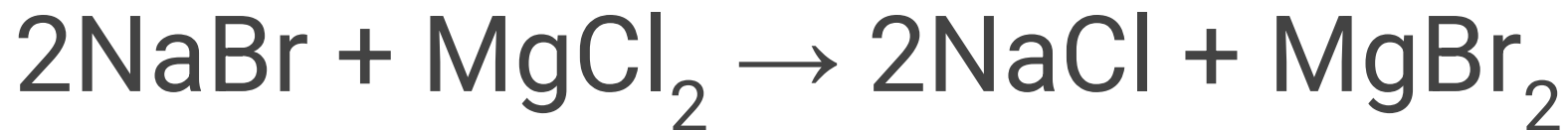
Predicting Reaction Types

- **Step 1:** Check whether the reaction is combustion. There are many telltale signs (CO_2 and H_2O are the products / the reactants are solely made of C, H, and O).
- **Step 2:** Only one product → *Synthesis*.
Only one reactant → Decomposition.
- **Step 3:** Check for the number of compounds.
 - Only 1 compound → Single displacement.
 - 2 compounds → Double displacement.



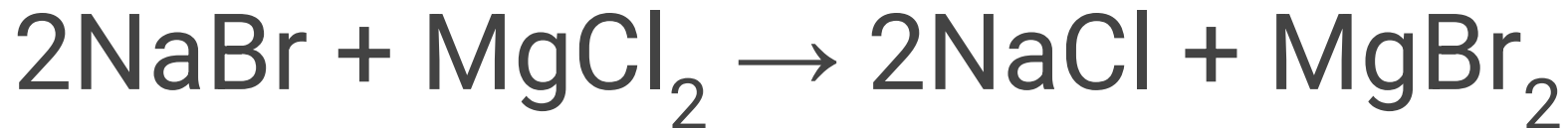
Example

What kind of reaction is this?



Example

What kind of reaction is this?

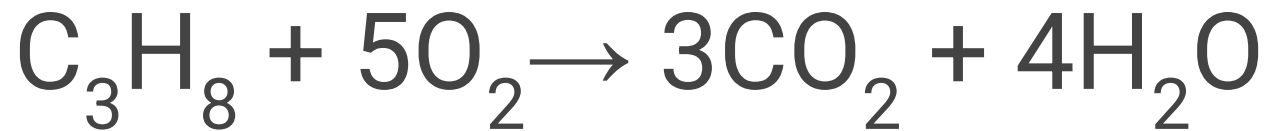


This is a **double displacement** reaction; Na and Mg are the cations that are being swapped here.



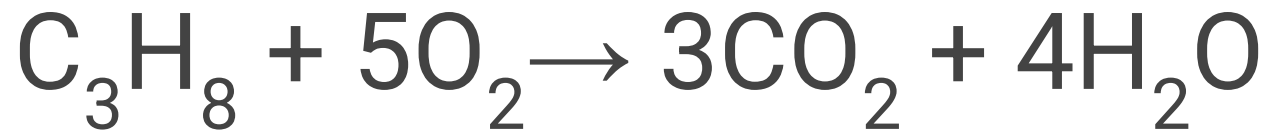
Example

What kind of reaction is this?



Example

What kind of reaction is this?



This reaction seems to feature oxygen as a reactant and CO_2 / H_2O as a product, so it's a **combustion** reaction.

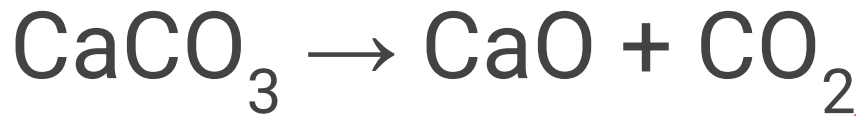


Predicting Decomposition

- There are quite a few rules to remember for decomposition.
- **Binary compounds** (made of two elements) break down into the natural states of both elements.

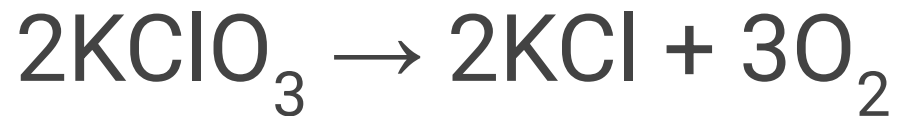


- **Carbonates** (containing CO_3^{2-}) break down into the metal oxide and CO_2 .
 - Metal oxide: Cation + oxygen



Predicting Decomposition

- **Chlorates** (containing ClO_3^-) break down into oxygen and *metal chloride* / a binary salt.
 - Binary salt $\rightarrow$ Compound with metal and chlorine.



Predicting Decomposition

- **Acids** form water (H_2O) and a metal oxide.



- Note: 1 Oxygen went to H_2O in this situation.

- **Bases** work similarly.



Example

Predict the products when $\text{Mg}(\text{ClO}_3)_2$ decomposes.



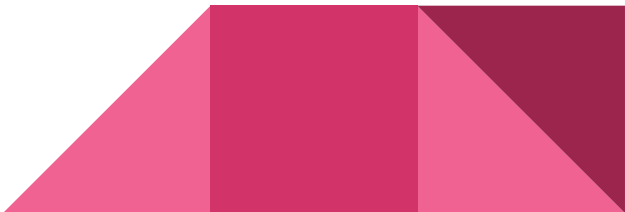
Example

Predict the products when $\text{Mg}(\text{ClO}_3)_2$ decomposes.

- The products are MgCl_2 and O_2 .



(balanced equation)



Example

Predict the products when $\text{Al}(\text{OH})_3$ decomposes. Is this an acid or a base?



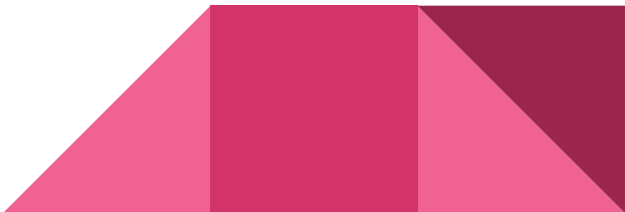
Example

Predict the products when $\text{Al}(\text{OH})_3$ decomposes. Is this an acid or a base?

- $\text{Al}(\text{OH})_3$ is a **base** because of its hydroxide ions, $(\text{OH})^-$.
- Products include Al_2O_3 and H_2O .



(balanced equation)



THANKS FOR ATTENDING

Please fill out this sign-in form before you leave today:



<http://tinyurl.com/JanCram24>

SCAN ME